GUIDE GLASS

INNOVATIVE DESIGN PROJECT REPORT

Submitted by

BABU R	**2303412510621030**
DHIVAKAR B	**2303412510621048**
KAMALESH V	**2303412510621083**

in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

in

ELECTRONICS AND COMMUNICATION ENGINEERING

SRI SAIRAM ENGINEERING COLLEGE

(An Autonomous Institution; Affiliated to Anna University, Chennai - 600 025)

ANNA UNIVERSITY: CHENNAI 600 025

MAY 2026

SRI SAIRAM ENGINEERING COLLEGE

(An Autonomous Institution; Affiliated to Anna University, Chennai -600 025)

BONAFIDE CERTIFICATE

Certified that this Innovative Design Project report **"GUIDE GLASS"** is the bonafide work of **"BABU R (2303412510621030), DHIVAKAR B (2303412510621048), KAMALESH V (2303412510621083)"** who carried out the **20ECPJ601 – INNOVATIVE DESIGN PROJECT** under my supervision.

SIGNATURE	SIGNATURE	SIGNATURE
Supervisor	Project Coordinator	Head of the Department
Mr. T. SIVASAKTHI	**Ms. K. SUBHASHINI**	**Dr. J. THAMIL SELVI**
Assistant Professor	**Associate Professor**	**Professor**
Department of ECE, Sri Sairam Engineering College (Autonomous), Chennai – 600 044	Department of ECE, Sri Sairam Engineering College (Autonomous), Chennai – 600 044	Department of ECE, Sri Sairam Engineering College (Autonomous), Chennai – 600 044

Submitted for project Viva – Voce Examination held on ________________________

INTERNAL EXAMINER **EXTERNALEXAMINER**

ACKNOWLEDGEMENT

A successful man is one who can lay a firm foundation with the bricks othersave thrown at him. — *David Brinkley*

Such a successful personality is our beloved founder Chairman, **Thiru. MJF. Ln. LEO MUTHU**. At first, we express our sincere gratitude to our beloved Chairman through prayers, who in the form of a guiding star has spread his wings of external support with immortal blessings.

We express our gratitude to our CEO **Dr. J. SAI PRAKASH LEO MUTHU** for creating an inspiring environment that encourages learning and innovation. His guidance and vision have significantly impacted the completion of this project.

We express our sincere thanks to our beloved Principal, **Dr. J. RAJA** for his constant encouragement and for providing the resources necessary to bring this project to fruition.

We are indebted to our Head of the Department, **Dr. J. THAMIL SELVI** for the insightful suggestions, mentorship, and continuous motivation, which have guided me at every stage.

We thank our Project Co-ordinator, **Ms.K. SUBHASHINI** who has been instrumental in coordinating efforts and ensuring smooth progress.

We express our gratitude and sincere thanks to our Guide **Mr.T. SIVASAKTHI** for his expertise, patience, and valuable insights. Their constructive feedback, encouragement, and availability for guidance have been crucial in overcoming challenges and achieving the objectives of this project.

We thank all the teaching and Non-teaching staff members of the Department of Electronics and Communication Engineering and all others who contributed directly or indirectly for the successful completion of the project.

ABSTRACT

Guide Glasses is an innovative project aimed at enhancing the mobility and independence of visually impaired individuals in crowded and dynamic environments. These smart glasses leverage cutting-edge technology, including computer vision, sensor integration, and AI-driven navigation, to assist users in navigating through unfamiliar or densely populated spaces. By providing real-time, contextual information about obstacles, potential hazards, and safe pathways, Guide Glasses empower visually impaired people to move freely and confidently without the need for constant assistance. The glasses utilize a combination of ultrasonic sensors, LIDAR, and cameras to detect and analyze the surrounding environment. Through this data, the glasses can identify obstacles such as pedestrians, walls, and furniture, alerting the wearer with auditory cues or haptic feedback. Additionally, Guide Glasses feature real-time navigation assistance, offering voice-guided instructions to help users reach their destinations while avoiding crowded or obstructed areas. The system can also adapt to different environments, from outdoor spaces with GPS - based navigation to complex indoor settings where crowd density and layout may vary. The project aims to address the challenges faced by visually impaired individuals in public spaces, promoting greater autonomy and safety. By combining advanced technology with intuitive user interfaces, Guide Glasses offer a seamless and user-friendly experience, allowing users to interact with the world around them in a more independent and confident manner. This innovation has the potential to significantly improve the quality of life for millions of people with visual impairments, contributing to a more inclusive and accessible society.

TABLE OF CONTENTS

LIST OF TABLES

LIST OF FIGURES

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AR	Augmented Reality
OCR	Optical Character Recognition

CHAPTER 1

EXECUTIVE SUMMARY

1.1 Overview of the Business Idea

The project focuses on developing smart eyeglasses for the visually impaired to improve their mobility and independence. The goal is to create advanced eyeglasses that integrate real-time obstacle detection, spatial awareness, and navigation assistance. The eyeglasses aim to enhance safety, provide better environmental awareness, and enable visually impaired individuals to navigate confidently in dynamic environments.

1.2 Vision & Mission

Vision: To enhance the independence and mobility of visually impaired individuals by providing an innovative wearable navigation solution.

Mission: To design smart eyeglasses that combine real-time obstacle detection and navigation technology, ensuring visually impaired individuals can navigate safely and independently.

1.3 Key Highlights

Target Market: Visually impaired individuals, assistive technology providers, rehabilitation centers, and organizations focused on disability inclusion. Unique Selling Proposition (USP): These smart eyeglasses will provide real-time navigation assistance and obstacle detection, offering a more efficient and accessible solution than traditional devices like canes.

CHAPTER 2

THE PROBLEM AND OPPORTUNITY

2.1 Problem Statement

Blind and visually impaired individuals face significant challenges in navigating their environment independently and safely. Current assistive technologies, such as canes, guide dogs, and basic wearable devices, have limitations in terms of range, real-time adaptability, and obstacle detection. While smart canes provide some assistance, they still fail to deliver comprehensive navigation solutions, especially in complex or dynamic environments. This creates a clear gap in the market for an advanced wearable solution, like smart eyeglasses, which can enhance real-time spatial awareness, obstacle detection, and overall mobility.

2.2 Market Opportunity

The assistive technology market in India for the blind is projected to grow substantially, with an estimated market size of ₹100 million by 2030. The rise in demand for independence, coupled with advancements in wearable technology, AI, and sensor integration, offers a significant opportunity to develop smart eyeglasses. These glasses could provide real-time feedback, helping blind individuals navigate more effectively, improving their independence and safety while addressing the current limitations of existing devices.

2.3 Problem Interviews and Survey Results

We conducted interviews with 5 visually impaired individuals, caregivers, and experts in assistive technology to identify the most pressing challenges and unmet needs:

1. 80% cited difficulty navigating through crowded or dynamic spaces as a major challenge.

2. 60% expressed dissatisfaction with existing assistive devices, stating that they lack realtime obstacle detection.
3. 40% mentioned that current wearable solutions are inaccurate, especially under varying lighting conditions.

CHAPTER 3

JUSTIFICATION FOR SDG GOAL

Primary SDG Goal: Reduced Inequalities

This project promotes equality by enhancing the mobility, independence, and social inclusion of visually impaired individuals. By breaking barriers to accessibility and providing opportunities for independent navigation, it contributes to reducing inequalities in everyday life.

Secondary SDG Goal: Industry, Innovation, and Infrastructure

The smart eyeglasses represent a breakthrough in assistive technologies, embodying innovation while contributing to the development of inclusive infrastructure. They drive advancements in accessibility, fostering collaboration between healthcare and tech industries.

Tertiary SDG Goal: Peace, Justice, and Strong Institutions

By empowering visually impaired individuals with greater mobility and independence, the project contributes to inclusive societies where people of all abilities can participate fully and equally. This fosters social justice and aligns with the vision of sustainable, inclusive communities.

CHAPTER 4

MARKET RESEARCH AND STRATEGY

3.1 Market Size Estimation

The market for Guide Glass—smart wearables designed for the visually impaired could reach between $500 million to $1.5 billion by 2026, with growth potential to $2-5 billion by 2030. This growth is driven by the rising global prevalence of visual impairment (affecting 285 million people), technological advancements in wearable devices, and increasing adoption of assistive technologies. While the initial market size is modest, the expanding demand for independent navigation solutions and the aging population present significant long-term opportunities for this niche sector

3.2 Target Audience and Customer Persona

- ·Visually Impaired and Blind Individuals
- ·Caregivers and Family Members
- ·Healthcare and Rehabilitation Professionals
- ·Tech-Savvy Consumers Interested in Wearables
- ·Non-Profit Organizations and NGOs

Customer persona:

Name: Sanjay

Age: 19

Background: College Student

Struggle: Uses a heavy and bulky design, faces general unease during long term wear

Need: More integrated model available at low cost

3.3 Industry Trends and Insights

The rising aging population and growing demand for telehealth and remote caregiving solutions create strong opportunities for assistive wearables. The trend toward augmented reality (AR) in consumer technology, along with expanding government and NGO support for disability aids, is accelerating innovation in this space. As manufacturing costs decrease, Guide Glass is expected to become more affordable and accessible to a broader audience, including seniors and visually impaired individuals worldwide.

3.4 Competition Analysis

- **Orcam**: Focused on Instant text recognition, Facial recognition, Product identification, AI powered, Discreet and wearable.
- **Iris vision**: Advance in Enhanced vision, Portability, Hands-free, Advanced AI powered, Personalized assistance
- **Maptic**: Based on Autonomous navigation, Precise location tracking, point of interest identification, Increased mobility

3.5 Unique Selling Proposition (USP)

The unique selling proposition (USP) of Guide Glass lies in its combination of advanced AIdriven navigation, real-time environmental feedback, and remote assistance capabilities—all seamlessly integrated into a wearable device. Unlike traditional assistive tools, Guide Glass provides visually impaired individuals with independent mobility by detecting obstacles, offering situational awareness, and guiding them safely through both familiar and unfamiliar environments. Additionally, the companion mobile app allows caregivers to track the user's location and provide real-time remote guidance, ensuring continuous support and peace of mind

CHAPTER 5

SOLUTION & PRODUCT DEVELOPMENT

5.1 Proposed Solution

The proposed solution for the problem of limited mobility and independence faced by visually impaired individuals is Guide Glasses, a smart wearable device designed to enhance navigation and safety. These glasses integrate advanced sensors, AI-driven navigation, and real-time environmental feedback to detect obstacles, provide situational awareness, and guide users through crowded and unfamiliar environments.

5.2 Product Features and Benefits

Low-light Image Enhancement: This model enhances the contrast of images in low-light conditions using a two-branch exposure-fusion network, allowing users to see better in dark environments.

Object Detection: It employs a transformer encoder-decoder model that can recognize 133 categories of objects, providing audio feedback to guide users about their surroundings.

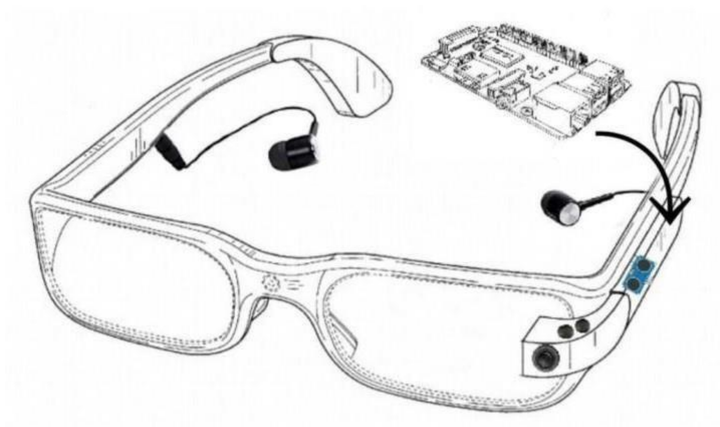
Salient Object Extraction: This feature helps in identifying and highlighting important objects in the user's environment, which is crucial for navigation and safety.

A. HC-SR04 Ultrasonic Sensor

Module	HC-SR04
Operating Voltage	5V DC
Operating Current	2mA
Effectual Angle	<15 degree
Ranging Distance	2 cm - 400 cm
Resolution	1 cm

B. Raspberry Pi 3 B+

Processor	Broadcom BCM2837B0, Cortex-A53 64-bit SoC @ 1.4GHz
Memory	1GB LPDDR2 SDRAM
Connectivity	2.4GHz and 5GHz IEEE 802.11.b/g/n/ac wireless LAN, Bluetooth 4.2, BLE Gigabit Ethernet over USB 2.0 (maximum throughput 300Mbps) 4 × USB 2.0 ports
Access	Extended 40-pin GPIO header
Video & sound	1 × full size HDMI MIPI DSI display port MIPI CSI camera port 4 pole stereo output and composite video port



Real-time Audio Feedback: The system provides users with information about surrounding objects through real-time audio output, enhancing their awareness and ability to navigate independently.

Tactile Display: In addition to audio feedback, the system includes a refreshable tactile display that allows users to perceive salient objects through their sense of touch, further aiding in navigation and understanding of their environment.

Client-Server Architecture: The system is designed with a client-server architecture, which helps in efficient data processing and communication between the smart glasses and a smartphone or server. This architecture supports features like streaming and load balancing, which are essential for real-time applications

Benefits:

- Enhanced Independence
- Real-Time Obstacle Detection
- Personalized Feedback
- Increased Safety
- Long Battery Life
- Stylish and Comfortable Design

5.3 Value Proposition Canvas

- **Customer Jobs:** Customers may report their issues to the company, offering feedback that could help improve future versions or updates of the glasses.
- **Pains:** Bulky or heavy designs can cause physical discomfort, including headaches, pressure on the nose or ears, or general unease during longterm wear.
- **Gains:** Simple, intuitive controls with minimal learning curves make users feel confident and in control. Hands free operation through voice commands or gestures would be a plus.

5.4 Minimum Viable Product (MVP)

The MVP aims to be a proof-of-concept smart glass design incorporating core features:

·Obstacle Detection and Navigation Assistance

·Low Light Image Enhancement

·Text Recognition (OCR) and Reading

·Voice-Controlled Interface and so on.

5.5 MVP Validation and Feedback

Validation will be done with Beta Testing with Target Audience. Feedback will be gathered from the peoples who got these disabilities (blind or visually impaired) and Healthcare Providers.

CHAPTER 6
BUSINESS MODEL AND STRATEGY

6.1 Business Model Overview

We will adopt a product sales and subscription-based model for the smart eyeglasses, targeting visually impaired individuals, healthcare providers, and assistive technology companies.

Revenue streams will include:

Product Sales: Sales of smart eyeglasses to individuals and organizations.

Subscription Fees: For access to premium features, such as real-time navigation updates and software enhancements.

Consultation Services: For integration with other assistive technologies and customized solutions for rehabilitation centers.

6.2 Revenue Model

Product sales and subscription fees will be the primary revenue sources, supplemented by consultation services. Additional income will come from partnerships with healthcare organizations, rehabilitation centers, and non- profit groups focused on accessibility, aiming to deploy the smart eyeglasses in large-scale assistive technology initiatives.

6.3 Marketing and Customer Acquisition Strategy

Branding: Position the smart eyeglasses as a revolutionary tool for independence and mobility, improving the quality of life for visually impaired individuals.

Partnerships: Collaborate with healthcare organizations, disability advocacy groups, and educational institutions to co-brand and expand the product's reach.

Retention: Offer customer support, software updates, and personalized features to ensure longterm satisfaction and adaptability in evolving environments

CHAPTER 7

FINANCIAL PLAN & PROJECTIONS

7.1 Financial Overview

Startup Costs : ₹2,00,000 (development, prototyping, initial operations)

Revenue Projections : ₹75,000 in Year 1, scaling to ₹7,00,000 by Year 5

7.2 Forecasted Profit & Loss (P&L)

Year1: Revenue: ₹75,000; Net Profit: ₹20,000

Year2: Revenue: ₹3,50,000; Net Profit: ₹1,20,000

7.3 Financial Projections

We expect to break even by Year 2, with growth and profitability accelerating in Year 3 as the customer base expands.

7.4 Unit Economics

Customer Acquisition Cost (CAC): ₹287

Customer Lifetime Value (CLV): ₹19,448

Payback Period: 18 months

7.5 Funding Plan and Capital Requirements

We are seeking ₹10,00,000 in seed funding for platform development, prototyping, and marketing for the first 18 months. Funds will be allocated as follows:

- 40% for prototyping and software development: ₹6,00,000 30% for marketing and customer acquisition: ₹2,00,000 30% for operational expenses: ₹2,00,000

CHAPTER 8

TEAM AND EXECUTION

8.1 Team Composition and Roles

CEO: Babu R –A third year ECE student contributing to technical aspects and implementation.

CTO: Dhivakar B– A third year ECE student managing coordination and planning.

COO: Kamalesh V– A third year ECE student overseeing operations and support.

8.2 Milestones & Execution Roadmap

- **Q1: Problem and Opportunity** – Define the problem within the Guide Glass from the customers and overcome the disabilities by giving standardized future models.

- **Q2: Market Research & Strategy, Solution & Product Development** – Conduct thorough market research to understand current solutions and identify areas for improvement. By giving better solution, the product will reach better market deals.

- **Q3: Prototype Development & Testing** – Develop a working prototype of the guide glass and perform live testing to assess performance. Conduct field testing for collect data for further optimization.

8.3 Risk Analysis and Mitigation Strategies

Risk:

System malfunctions or hardware failures could compromise user safety.

Mitigation:

- Conduct extensive testing under various conditions.
- Use high-quality, durable components and implement redundant safety mechanisms.
- Regular software updates and fail-safe features like alerting users in case of malfunction

Risk:

Challenges in AI accuracy for obstacle detection and object recognition in dynamic environments

Mitigation:

Train AI models with diverse datasets to handle complex scenarios. Incorporate continuous learning mechanisms for system improvement.

Risk:

Resistance to adopting new technology due to complexity or lack of trust.

Mitigation:

- Design a user-friendly interface with simple controls.
- Conduct educational campaigns, training sessions, and hands-on demonstrations.
- Partner with advocacy groups to build trust among the visually impaired community.

CHAPTER 9

CONCLUSION AND NEXT STEPS

9.1 SUMMARY OF KEY POINTS

- Our project focuses on developing smart eyeglasses to assist visually impaired individuals in navigating their environment with ease and independence.
- The eyeglasses will include AI-powered obstacle detection, real-time spatial awareness, and GPS integration.

Designed to be lightweight, ergonomic, and user-friendly, ensuring comfort for daily wear. ⬥ Emphasize affordability by partnering with NGOs, healthcare providers, and disability- focused organizations for wider accessibility.

- Focus on iterative user testing to enhance functionality and design, with a durable, modular structure for long-term use.
- Implement rigorous testing, comply with safety standards, and ensure robust data protection and user privacy.

9.2 FUTURE SCOPE

Prototype Development: Finalize the eyeglass design and develop the first working prototype. **Testing & Optimization:** Conduct both lab-based and real-world testing to refine the product's performance and user experience.

Scaling & Deployment: Prepare for scaling the production process and build partnerships for mass deployment through healthcare organizations and rehabilitation centers.

9.3 REFLECTION THE LIVE IN LAB EXPERIENCE

The Live-in-Lab experience for our project Guide Glass has been one of the most insightful and impactful parts of our academic journey. The opportunity to step outside the traditional classroom setting and immerse ourselves in a real-world environment enabled us to apply our theoretical knowledge to practical, socially relevant challenges.

Guide Glass—a wearable assistive device designed to support visually impaired individuals in navigation and daily tasks—was a concept that demanded not just technological innovation, but also a deep understanding of the end users' lives. Through the Live-in-Lab initiative, we were able to directly interact with visually impaired individuals, observe their daily routines, and gain first-hand knowledge of the limitations they face with existing assistive technologies.

One of the most valuable aspects of the experience was user feedback. Early prototypes of Guide Glass were tested in real-life scenarios, and the feedback we received was instrumental in improving the design, interface, and overall functionality of the product. For example, we discovered that audio feedback needed to be clearer in noisy environments and that users preferred voice-command features to be simple and intuitive. These insights helped us make significant improvements that would not have been possible in a purely lab-based setting.

Working closely with communities and stakeholders gave us a more holistic perspective on engineering design—emphasizing empathy, adaptability, and user-centric thinking. We also learned the importance of interdisciplinary collaboration, as the project required inputs from software developers, designers, and healthcare professionals. This collaborative spirit enriched our learning process and taught us how to navigate real-world challenges such as budget constraints, field testing, and ethical considerations.

The Live-in-Lab experience also helped us develop essential soft skills—communication, teamwork, and project management—that are crucial for any professional career. Presenting our project to community members and gathering their suggestions encouraged us to think critically and iterate rapidly based on their needs.

In conclusion, the Live-in-Lab journey transformed Guide Glass from a conceptual idea into a practical, impactful solution. It bridged the gap between innovation and implementation, and instilled in us a sense of social responsibility as future engineers. We are grateful for the opportunity and believe that such immersive learning experiences are vital in creating solutions that truly serve society.

REFERENCES

1. Azameti, A., Katsriku, F., Owusu, E., & Abdulai, J.-D. (2024). Improving Accessibility for the Visually Impaired: A Review of Assistive Technologies and Future Prospects.
2. Katare, K. K., Yousaf, I., & Lau, B. K. (2024). Challenges and Solutions in Wearable Assistive Devices for the Blind: Focus on Smart Glasses.
3. Ameen, A. M., Ahmed, M. I., Elsadek, H., & Anis, W. R. (2024). Real-Time Obstacle Detection for Smart Glasses: A Wearable Solution for Visually Impaired Individuals.
4. IEEE Xplore Digital Library. (2023). Smart Eyewear for the Visually Impaired: An Overview of Current Trends and Future Research Directions.
5. Gupta, A. (2024). Advancements in Smart Wearables for the Disabled: Focusing on Mobility and Navigation Solutions.
6. Sharma, B., et al. (2023). Design and Simulation of Smart Glasses for Navigation Assistance for the Visually Impaired. Journal of Assistive Technology Research.
7. Assistive Technology for Visually Impaired Individuals: A Survey on Smart Wearables. International Journal of Assistive Technology, 2023.
8. Wearable Technology and IoT for Blind Navigation: Current Applications and Future Prospects. International Journal of Human-Computer Interaction, 2024.

APPENDIX I

FIRST LEVEL SIMULATION AND OUTPUT

```
import cv2
from ultralytics import YOLO
import pyttsx3
import time
modelA = YOLO("yolo11n.pt")
modelB = YOLO("best.pt")
engine = pyttsx3.init()
engine.setProperty('rate', 160)
last_alert = ""
last_time = 0
cooldown = 3  # seconds
cap = cv2.VideoCapture(0)
while True:
    ret, frame = cap.read()
    if not ret:
        break
    resultsA = modelA(frame, conf=0.25, verbose=False)
    resultsB = modelB(frame, conf=0.25, verbose=False)

    annotated = frame.copy()
    for r in resultsA:
        for box in r.boxes:
            x1,y1,x2,y2 = map(int, box.xyxy[0])
            label = modelA.names[int(box.cls)]
            cv2.rectangle(annotated, (x1,y1), (x2,y2), (0,255,0), 2)
            cv2.putText(annotated, f"A:{label}", (x1,y1-10),
                        cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0,255,0), 2)
    for r in resultsB:
        for box in r.boxes:
            x1,y1,x2,y2 = map(int, box.xyxy[0])
            label = modelB.names[int(box.cls)]
            cv2.rectangle(annotated, (x1,y1), (x2,y2), (255,0,0), 2)
            cv2.putText(annotated, f"B:{label}", (x1,y1-10),
                        cv2.FONT_HERSHEY_SIMPLEX, 0.7, (255,0,0), 2)
            current_time = time.time()
            alert = f"{label} detected"
            if alert != last_alert or current_time - last_time > cooldown:
                engine.say(alert)
                engine.runAndWait()
                last_alert = alert
                last_time = current_time
            break  # speak only once per frame

    cv2.imshow("YOLO Dual-Model Detection", annotated)

    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```

Fig A.1 ultralytics for python

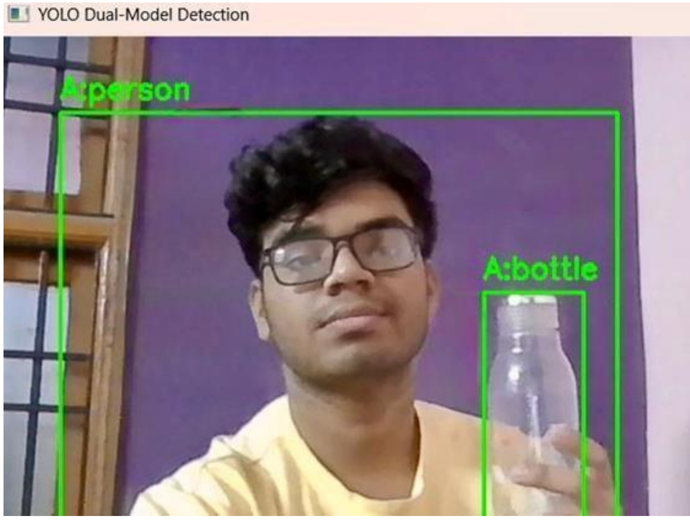


Fig A.1.2 Detection of bottle **Fig A.1.3 Detection of cell phone**

The dual YOLO-based real-time object detection system is an intelligent computer vision application that uses two deep learning YOLO models to detect and recognize objects from live camera input while simultaneously providing voice alerts through a text-to-speech mechanism. The system operates on the principle of single-stage object detection, where each video frame is processed in real time to predict object classes, confidence scores, and bounding box locations in a single forward pass of a convolutional neural network, enabling high-speed performance. One model is used for general object detection, while the second model is custom-trained for specific objects, thereby improving detection coverage and accuracy. Detected objects are visually highlighted using bounding boxes and labels, and their names are converted into spoken alerts to assist users, especially in accessibility-oriented applications. To avoid repetitive announcements, a time-based cooldown mechanism is employed so that alerts are generated only for new detections or after a specified interval. By integrating real-time video acquisition, deep learning-based object detection, visual annotation, and audio feedback, the system provides an efficient, user-friendly, and scalable solution suitable for assistive technologies, surveillance systems, and autonomous applications.

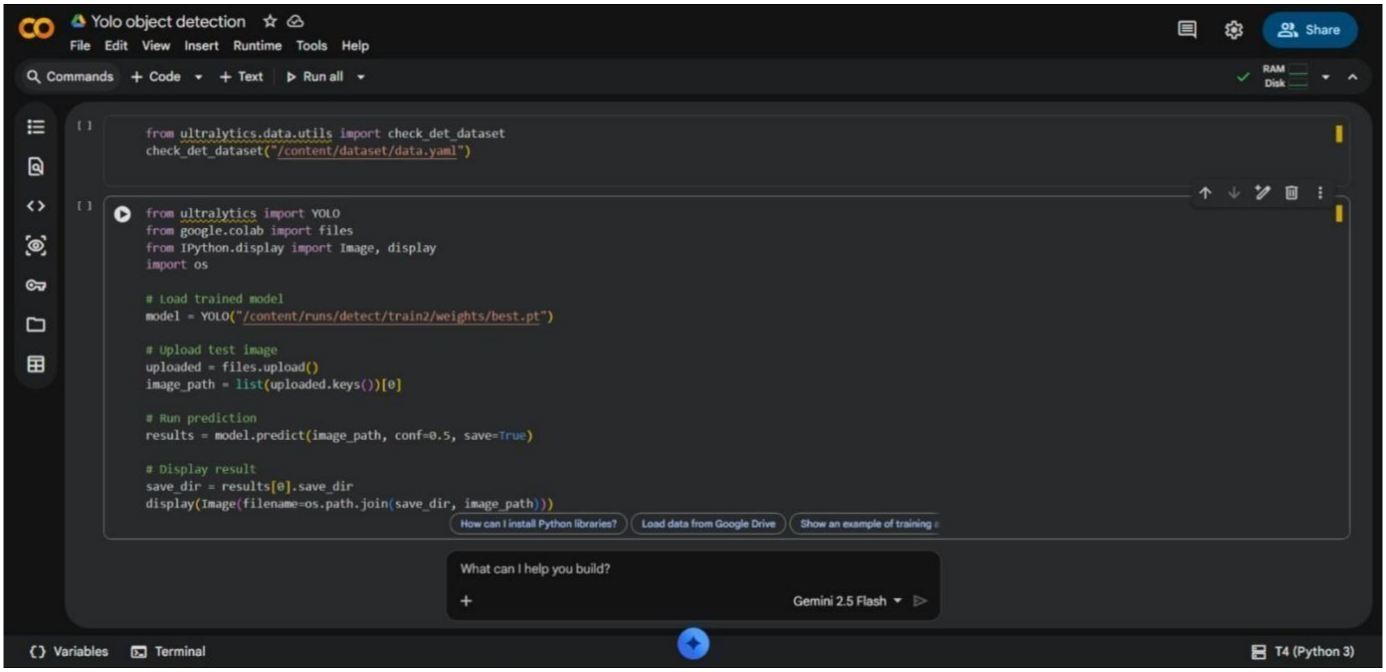
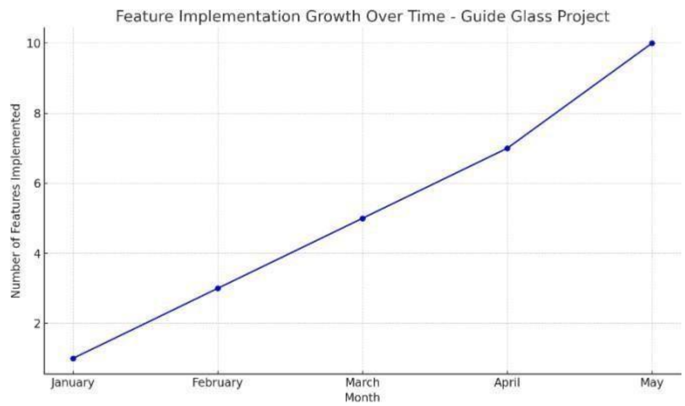


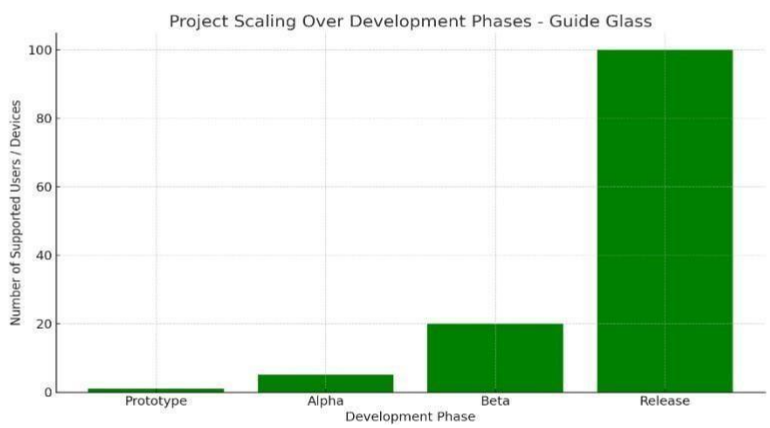
Fig A.1.3 Google colab code for YOLO

This Google Colab–based YOLO object detection program demonstrates the post-training inference stage of a deep learning object detection system, where a previously trained YOLO model is used to identify objects in a test image. The system first verifies the integrity of the custom detection dataset configuration to ensure correct class and path definitions, which is essential for reliable model operation. A trained YOLO model is then loaded from stored weights, representing the learned features and object representations obtained during training. The user uploads a test image through the Colab interface, which enables interactive evaluation without local system dependencies. The model processes the image using single-stage object detection, predicting object classes, confidence scores, and bounding box locations in one forward pass of the neural network. A confidence threshold is applied to filter weak detections, and the resulting annotated image is automatically saved and displayed. This approach highlights the effectiveness of cloud-based platforms like Google Colab for rapid testing, visualization, and validation of deep learning object detection models without requiring real-time video input or complex hardware setup.

PROJECTION SCALING



The graph shows a steady increase in the number of features added to the Guide Glass project over time. Early growth was slow due to setup and core development. As the team gained experience and foundational systems were completed, development accelerated. This reflects an iterative or agile approach, where each phase built on the last. The sharp rise in later months indicates improved efficiency and rapid integration of advanced features.

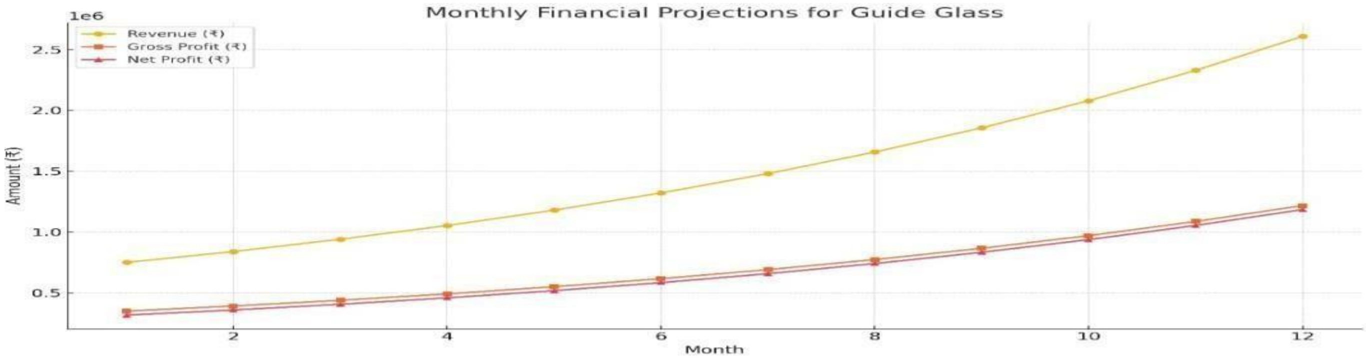


The scaling graph shows how the Guide Glass project progressed from a single-user prototype to supporting a larger user base. As the system matured through Alpha, Beta, and Release phases, its capacity to handle more users/devices increased. This growth reflects improvements in hardware reliability, software optimization, and user adaptability, demonstrating the project's potential for real-world deployment at scale.

Basic Guide Glass Hardware Code (Arduino C++)

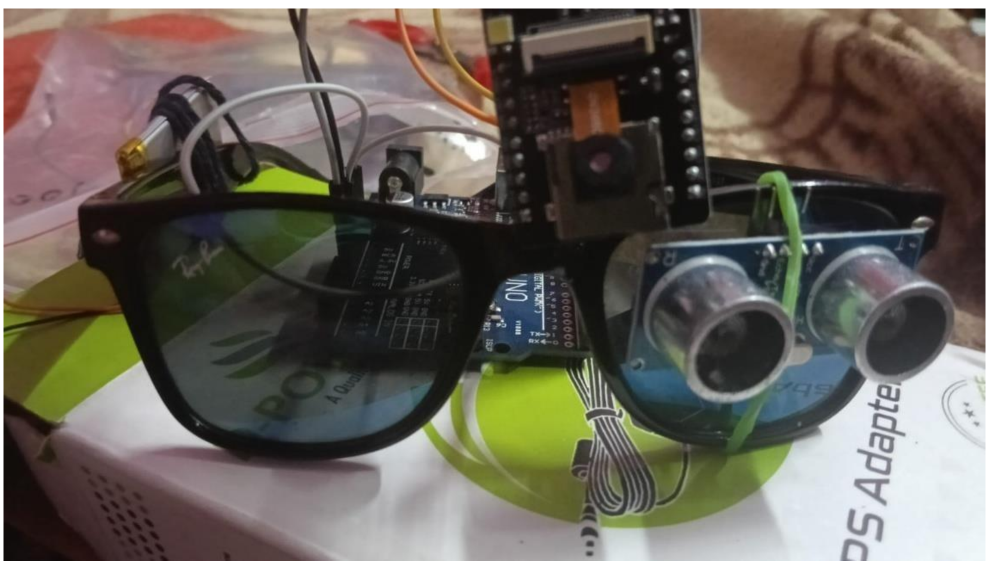
```
#include <Wire.h>

 #include <Adafruit_SSD1306.h>

#include <Adafruit_GFX.h>

#include <TinyGPS++.h>

#include <HardwareSerial.h>

// OLED setup

#define SCREEN_WIDTH 128

 #define SCREEN_HEIGHT 64 Adafruit_SSD1306 display(SCREEN_WIDTH,
SCREEN_HEIGHT, &Wire, -1);

// GPS setup TinyGPSPlus gps; HardwareSerial GPS(1);

void setup() {

 Serial.begin(115200);

 GPS.begin(9600, SERIAL_8N1, 16, 17); // RX, TX for ESP
```

Features	Raspberry Pi 3	Raspberry pi zero w
Size & price	It has a height of 2.22 inches and width of 3.37 inches 35$	It has a height of 1.18 inches and width of 2.55 inches 10$
USB	It has four USB 2.0 ports	It only has one Micro USB
Wi-Fi & Bluetooth	Both has wireless capabilities, 2.4GHz 802.11n wireless LAN and Bluetooth Classic 4.1	
Ethernet	10/100 Mbit/s	No Ethernet connectivity
Analog Video	Available access	
LCD	Available	Not Available
Camera	Available connections	
GPIO (General Purpose Input/Output)	Both has 40 GPIO pins Both allow the usage of +3.3V, +5V, ground, I2C, and SPI	
Speed	It is 20% faster than the Raspberry pi zero w	It is slower than Raspberry pi 3
Core	Quad core	Single core
GPU	Both use Broadcom VideoCore IV	
Memory (SDRAM)	It has a 1GB memory	It has a 512 MB memory



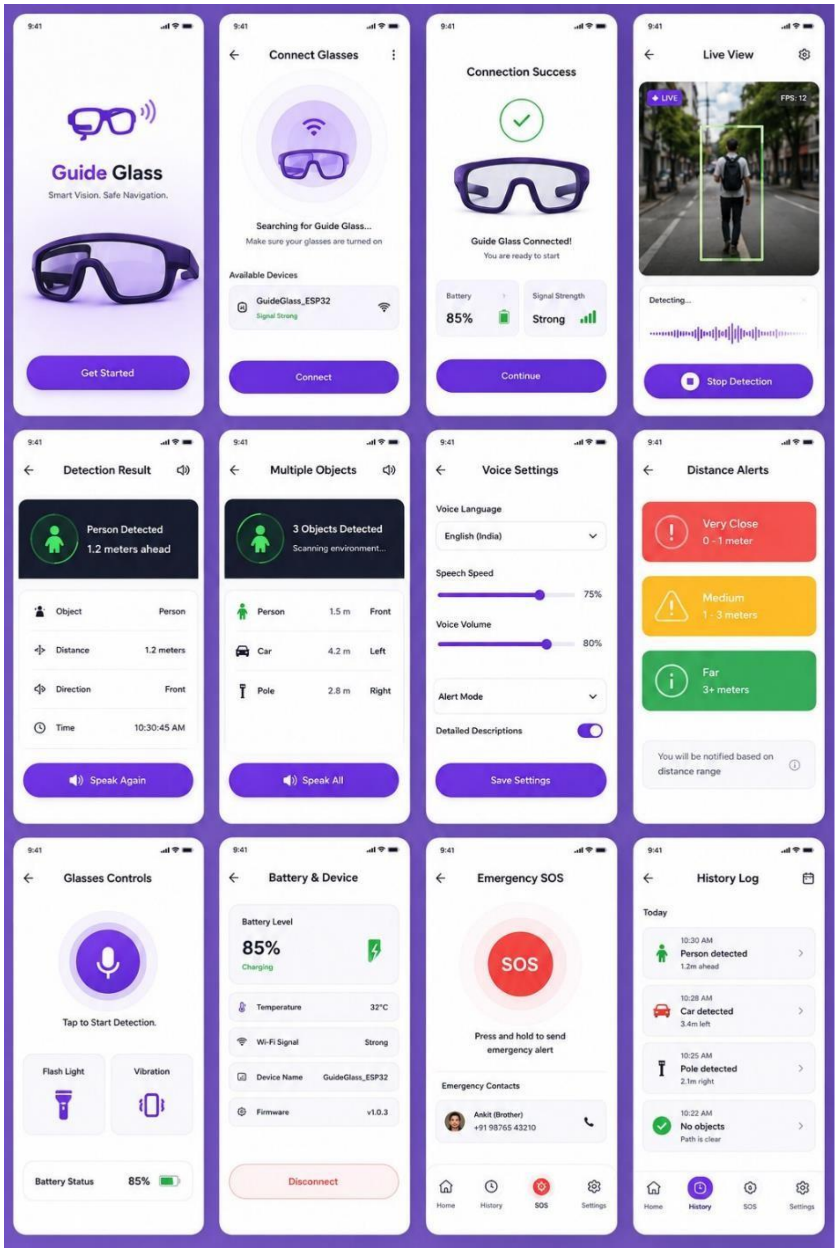
The graph presents a 12-month financial projection showcasing Revenue, Gross Profit, and Net Profit. These are core metrics that reflect the financial health and growth of a product-based business.

Guide Glass Hardware Prototype



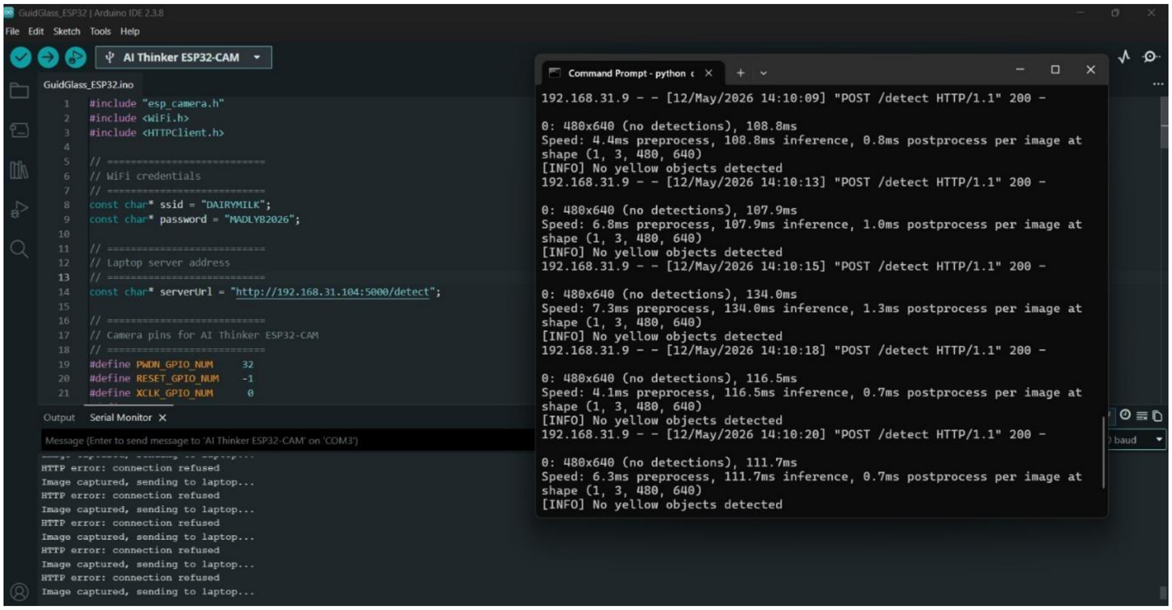
This image shows the physical hardware prototype of the Guide Glass project. It consists of an ESP32-CAM mounted on eyeglasses for real-time image capture and an ultrasonic sensor for obstacle distance detection. The system provides audio feedback to visually impaired users, helping them identify objects and avoid obstacles safely while navigating.

Guide Glass Mobile Application UI Design



This image displays the mobile application designed for Guide Glass. The app allows users to connect the smart glasses, monitor live camera detection, receive voice-based object alerts, adjust voice settings, check battery status, and access emergency SOS features. The user-friendly interface enhances accessibility and makes the system easier to control.

Arduino IDE + Detection Server Running



This image shows the software setup of the Guide Glass project. The ESP32-CAM is programmed using Arduino IDE to capture images and send them to a local AI detection server running on a laptop. The Python server processes each frame using object detection models and returns the detected object information. This confirms successful communication between the smart glasses hardware and the AI processing system.

ESP32-CAM Programming for Real-Time Image Acquisition and AI-Based Object Detection in Guide Glass

```
#include "esp_camera.h"
#include <WiFi.h>
#include <HTTPClient.h>
const char* ssid = "DAIRYMILK";
const char* password = "MADLYB2026";
const char* serverUrl = "http://192.168.31.104:5000/detect";
#define PWDN_GPIO_NUM     32
#define RESET_GPIO_NUM    -1
#define XCLK_GPIO_NUM      0
#define SIOD_GPIO_NUM     26
#define SIOC_GPIO_NUM     27
#define Y9_GPIO_NUM       35
#define Y8_GPIO_NUM       34
#define Y7_GPIO_NUM       39
#define Y6_GPIO_NUM       36
#define Y5_GPIO_NUM       21
#define Y4_GPIO_NUM       19
#define Y3_GPIO_NUM       18
#define Y2_GPIO_NUM        5
```

```
#define VSYNC_GPIO_NUM     25
#define HREF_GPIO_NUM      23
#define PCLK_GPIO_NUM      22

void setup() {
  Serial.begin(115200);
  Serial.println("Guide Glass Starting...");
  camera_config_t config;
  config.ledc_channel = LEDC_CHANNEL_0;
  config.ledc_timer   = LEDC_TIMER_0;
  config.pin_d0       = Y2_GPIO_NUM;
  config.pin_d1       = Y3_GPIO_NUM;
  config.pin_d2       = Y4_GPIO_NUM;
  config.pin_d3       = Y5_GPIO_NUM;
  config.pin_d4       = Y6_GPIO_NUM;
  config.pin_d5       = Y7_GPIO_NUM;
  config.pin_d6       = Y8_GPIO_NUM;
  config.pin_d7       = Y9_GPIO_NUM;
  config.pin_xclk     = XCLK_GPIO_NUM;
  config.pin_pclk     = PCLK_GPIO_NUM;
  config.pin_vsync    = VSYNC_GPIO_NUM;
  config.pin_href     = HREF_GPIO_NUM;
  config.pin_sscb_sda = SIOD_GPIO_NUM;
  config.pin_sscb_scl = SIOC_GPIO_NUM;
  config.pin_pwdn     = PWDN_GPIO_NUM;
  config.pin_reset    = RESET_GPIO_NUM;
  config.xclk_freq_hz = 20000000;
  config.pixel_format = PIXFORMAT_JPEG;
  config.frame_size   = FRAMESIZE_QVGA;  // 320x240
  config.jpeg_quality = 12;
  config.fb_count     = 1;
  esp_err_t err = esp_camera_init(&config);
  if (err != ESP_OK) {
    Serial.printf("Camera init failed: 0x%x\n", err);
    return;
  }
  Serial.println("Camera Ready!");

  WiFi.begin(ssid, password);
  Serial.print("WiFi connecting");
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
  Serial.println("");
  Serial.println("WiFi connected!");
  Serial.print("ESP32 IP: ");
  Serial.println(WiFi.localIP());
  Serial.println("Sending images to laptop for detection...");
```

```
}

void loop() {
  if (WiFi.status() != WL_CONNECTED) {
    Serial.println("WiFi lost, reconnecting...");
    WiFi.begin(ssid, password);
    delay(3000);
    return;
  }

  camera_fb_t* fb = esp_camera_fb_get();
  if (!fb) {
    Serial.println("Camera capture failed");
    delay(1000);
    return;
  }
  Serial.println("Image captured, sending to laptop...");

  HTTPClient http;
  http.begin(serverUrl);
  http.addHeader("Content-Type", "image/jpeg");
  int httpCode = http.POST(fb->buf, fb->len);

  if (httpCode > 0) {
    String response = http.getString();
    Serial.println("Server response: " + response);
  } else {
    Serial.printf("HTTP error: %s\n", http.errorToString(httpCode).c_str());
  }
  http.end();
  esp_camera_fb_return(fb);
  delay(2000); // Send image every 2 seconds
}
```

The ESP32-CAM module serves as the core image-capturing unit of the Guide Glass system. It is programmed using Arduino IDE to initialize the onboard camera and establish a wireless connection with a local Wi-Fi network. Once connected, the device continuously captures images from the user's surroundings and transmits them to a laptop-based AI server through HTTP communication.

The transmitted images are processed using an object detection algorithm running on the server, which identifies nearby objects and obstacles. The detection results are then sent back to the system, enabling the Guide Glass to provide real-time voice feedback to assist visually impaired users in navigation and obstacle avoidance.

The program includes camera configuration, Wi-Fi connection management, image acquisition, data transmission, and error-handling mechanisms to ensure reliable performance. By combining embedded hardware with AI-based image processing, this system creates an intelligent wearable assistive device for safer and more independent mobility.

APPENDIX III

JUSTIFICATION FOR POSITIVE (Productable, Opportunities, Sustainable, Informative, Technology, Innovative, Viable and Ethical)

S.No	PARAMETERS	JUSTIFICATION	RATING (1 to 5)
1.	Productable	The device can be easily manufactured with widely available sensors, motors, and controllers, allowing for mass production.	
2.	Opportunities	There's a growing need for affordable, portable healthcare devices in rural areas, offering opportunities to improve Glaucoma detection and treatment	
3.	Sustainable	The device uses low power consumption, making it energy-efficient and suitable for long-term use in various environments.	
4.	Informative	It offers real-time intraocular pressure (IOP) data, which is crucial for doctors and patients to track the progression of Glaucoma and adjust treatment.	
5.	Technology	It offers real-time intraocular pressure (IOP) data, which is crucial for doctors and patients to track the progression of Glaucoma and adjust treatment.	
6.	Innovative	By combining air-puff technology with dual-eye monitoring, it offers an innovative approach to Glaucoma management that enhances accuracy and patient comfort.	
7.	Viable	With affordable and accessible components, the device is feasible for mass production and deployment, making it viable for healthcare systems in both urban and rural settings.	
8.	Ethical	The project is designed to enhance accessibility to eye care, ensuring early diagnosis and reducing the need for invasive procedures, benefiting underserved populations.	

Imagine the Future and

Make it happen!

Together let's build a better world where there is NO POVERTY and ZERO HUNGER.
We have GOOD HEALTH AND WELL BEING QUALITY EDUCATION and full GENDER EQUALITY everywhere.
There is CLEAN WATER AND SANITATION for everyone. AFFORDABLE AND CLEAN ENERGY
which will help to create DECENT WORK AND ECONOMIC GROWTH. Our prosperity shall be fuelled
by investments in INDUSTRY, INNOVATION AND INFRASTRUCTURE that will help us to
REDUCE INEQUALITIES by all means. We will live in SUSTAINABLE CITIES AND COMMUNITIES.
RESPONSIBLE CONSUMPTION AND PRODUCTION will help in healing our planet.
CLIMATE ACTION will reduce global warming and we will have abundant,
flourishing LIFE BELOW WATER, rich and diverse LIFE ON LAND.
We will enjoy PEACE AND JUSTICE through STRONG INSTITUTIONS
and will build long term PARTNERSHIPS FOR THE GOALS.

For the goals to be reached,
everyone needs to do their part:
governments, the private sector,
civil society and **People like you.**

Together we can...

Sai Prakash Leo Muthu

Chairman & CEO – Sairam Institutions